



5965 - An Archival Study of Cosmic Transients in Existing JWST Observations

Cycle: 3, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

The James Webb Space Telescope (JWST) is a prolific transient factory. It enables time-domain astronomy to magnitudes and wavelengths beyond the reach of any other current and future transient survey programs. We propose to conduct a comprehensive analysis of a set of archival NIRCам observations with multi-epoch coverage and the development of a transient detection pipeline customized for the NIRCам data. The entire dataset consists of 49 fields covering ~248 square arcminutes with a depth fainter than the 28th mag. These archival observations hold abundant uncovered transients with diverse physical characteristics, including lensed stars, high redshift supernovae, and Active Galactic Nuclei (AGNs) with photometric variability. These transients are a unique asset for the JWST, which can be used to study the formation and evolution of stars in the early

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universe, and set constraints on cosmological parameters. Through this project, we will develop a data processing pipeline designed to discover and classify transients with NIRCам data. The JWST has a complicated PSF and different detector orientations at different epochs. We will apply three commonly used image differencing packages, HOTPANTS, ZOGY, and SFFT, and compare their performances. We will use the NIRCам data to retrain our neural network-based candidate screening software for automatic spurious source rejection and light curve classification. Our pipeline and the transient catalog will be publicly accessible to the JWST community. Our team has a special interest in using these transients to derive the supernova rates and use them to constrain the initial mass function and star formation rates at high redshifts.

OBSERVING DESCRIPTION